

Neural Network models (80-20 split): Metrics, Voltage and Power plots

Here, we use 3 neural network models for voltage and power comparison:

1. Dense Neural Network (DNN)
2. Deep Neural Network (DeepNN)
3. Long Short Term Memory (LSTM)

Following are the metrics:

Voltage Prediction:

Model	MAE	RMSE	R ²	Remarks
DNN	0.0684	0.0989	0.8804	Baseline performance
DeepNN	0.0721	0.0944	0.8911	Slight improvement in R ²
LSTM	0.0645	0.0875	0.9064	Best voltage model

Power Prediction:

Model	MAE	RMSE	R ²	Remarks
DNN	1.2917	1.9502	0.99918	Strong baseline
DeepNN	0.9059	1.5187	0.99950	Best power model
LSTM	4.8755	9.5607	0.98030	Temporal model not beneficial

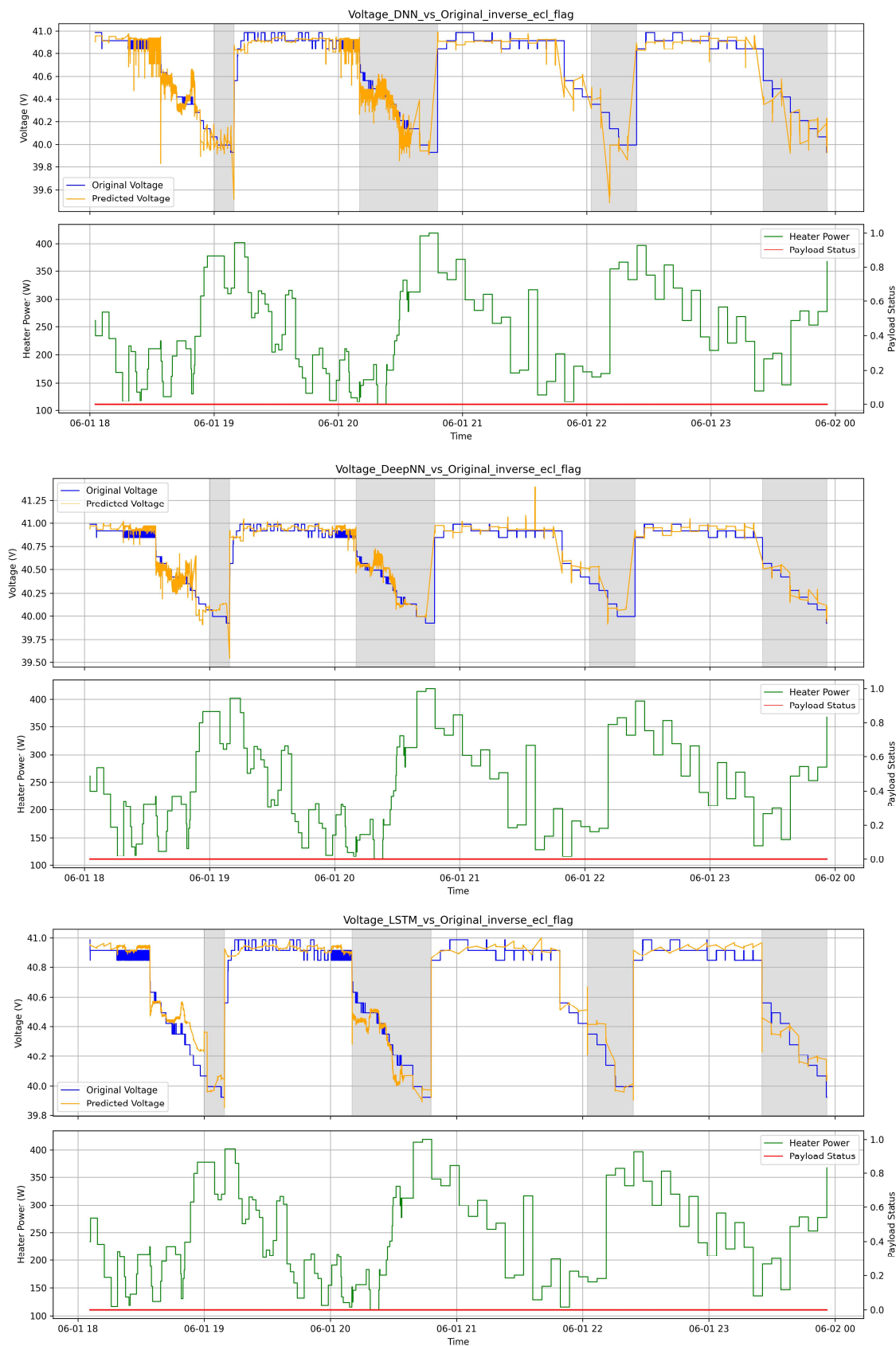
Voltage:

- Past states influence present voltage
- Shows moderate temporal dependence
- Therefore LSTM performs best

Power:

- Power dominated by instantaneous load and heater activity
- Minimal historical memory effect
- Shows weak temporal dependence
- Therefore feedforward DeepNN performs best

Voltage Graph Plots:



Power graph plots:

